

Assignment 1:

Q1: In your own words, explain what diabetic retinopathy is and why it is a serious public health problem. Think about: Who gets it? What does it do to the eye? What happens if it is not caught in time? Why can't we just send every patient to a specialist?

A1: Diabetic retinopathy is an eye condition which results from high blood sugar levels, which are most commonly found in diabetic patients. These high levels of blood sugar slowly damage the blood vessels in the retina. This is a serious problem, since about 500 million people worldwide live with some sort of diabetic disease, which could lead to conditions like diabetic retinopathy. If not treated immediately, diabetic retinopathy could lead to permanent blindness. The problem with treating this disease is that there simply aren't enough specialists (ophthalmologists) who are capable of treating patients.

Q2: AI needs to LEARN before it can diagnose. Explain, in simple terms, what 'training data' is and why it was important that 54 ophthalmologists labeled 128,000 photos for Google's AI. What would happen if the AI was trained only on photos from one country or one ethnic group?

A2: Training data is simple forms of data that is given to an AI model for training. The model trains on this "example data" and learns from it, applying what it learned to future data. The ophthalmologists who had labeled the photos for Google's AI had done so to increase robustness and minimize bias. If the AI were to be trained on just one photo from one country/ethnic group, then the model would be biased towards that group, and would label any different group as abnormal.

Q3: Google's retina AI achieved 90% sensitivity and 98% specificity in trials. Using the Cheat Sheet to understand these terms, explain in plain English what these numbers mean for a patient. Is 90% sensitivity good enough? What happens to the 10% of cases it misses?

A3: 90% sensitivity means that 90% of the time, the model was able to correctly identify a real case of diabetic retinopathy. 98% specificity means that only 2% of patients were falsely diagnosed with diabetic retinopathy. Ideally, a 100% sensitivity would be best for any scenario. The missing 10% of cases are incorrectly labeled as disease-free, which means that they actually have diabetes but were told they were healthy.

Q4: The AI was deployed in Thailand's rural clinics where no eye doctor was available. Describe TWO specific benefits this created for patients and ONE challenge or risk that came with using AI in this setting. Consider: privacy, equipment availability, internet access, patient trust.

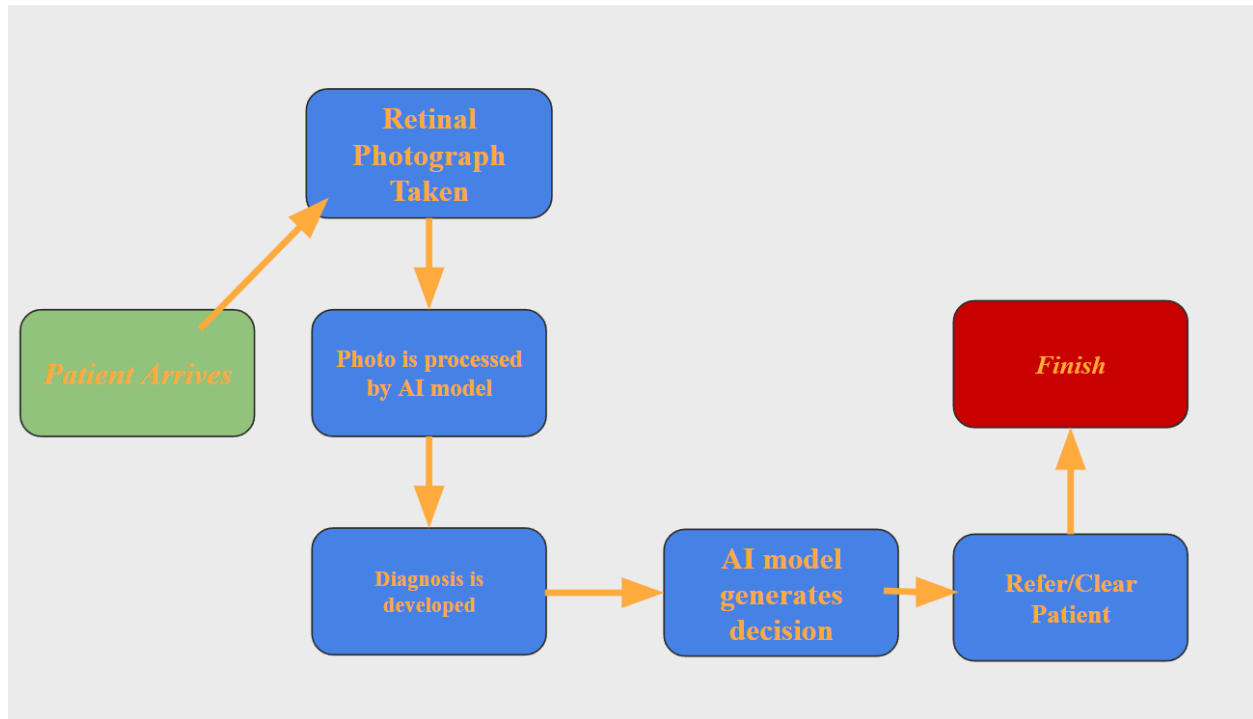
A4: Patients in the rural areas wouldn't have to wait for a specialist to diagnose them, since the AI would diagnose them instead. Patients also gain access to an alternate way to diagnose diseases, along with a human specialist. However, there is a challenge to the use of AI in this setting. Internet access is limited in these areas, so actually accessing the AI for diagnosis will be difficult if locals do not have access to technology.

Q5: Create a simple comparison: list TWO things the AI can do BETTER than a human doctor in this scenario, and TWO things the human doctor can do that the AI CANNOT. Conclude: do you think this AI should be used without any doctor involvement? Why or why not?

A5: AI can diagnose multiple patients much quicker than a human doctor, and can essentially be deployed anywhere where there is internet access. But, a human doctor can make ethical decisions and form a human connection with the patient, which is something that an AI model cannot do. I believe that the AI should at least be monitored by a human doctor, as a failsafe. If the model fails to do its job, the human doctor can take over and help the patient with their needs.

Q6: Draw a simple flowchart (on paper or digitally) showing the journey of one diabetic patient: from visiting a rural clinic, getting their retina photograph taken, through the AI analysis process, to either 'referred to specialist' or 'no disease detected, recheck in 12 months.' Label at least 5 steps. Attach or photograph your diagram.

A6:



Ethics Q1: Privacy Concern: The AI needs photographs of patients' eyes to work. These photographs are medical data. Who should own this data — the patient, the hospital, the AI company (Google), or the government? What could go wrong if this data was misused?

Ethics A1: Realistically, the patient and the hospital should be the only ones who own the data, but the hospital should only gain ownership if the patient allows it. If the data were to be misused, this could lead to privacy breaches and security flaws, like the patient's identity being released to the public unexpectedly.

Ethics Q2: The Equity Question: Currently, the AI works best on patients whose data it was trained on. If an AI company trains mostly on patients from wealthy countries, the AI may be less accurate for patients in Africa or Asia. Is this fair? Whose responsibility is it to fix this — the company, the government, or doctors?

Ethics A2: This is not a fair approach. The company holds responsibility for this. They should train the model on every type of patient, whether they'd be from a wealthy country, a poor country, or a developing country.

Ethics Q3: Trust and Transparency: Imagine you are a patient in a rural clinic. A nurse tells you: 'A computer analyzed your eye photo and says you don't have disease — you can go home.' How would you feel? What information would you want to know before trusting this result?

Ethics A3: Personally, I would want a human doctor to analyze the photo or do any other tests themselves rather than having an AI model diagnose me. I would want to know for sure if I have the disease or not, whether it be a doctor confirming the results of the AI diagnosis or both the AI's and doctor's separate diagnoses.

Assignment 2:

Q1: Explain in your own words why a regular TV weather forecast is NOT useful enough for a farmer like the Torres family. What specific information does a farmer need that a general forecast doesn't provide? Give at least THREE specific examples of farming decisions that depend on precise local weather data.

A1: TV weather forecasts are too general for farmers, as they do not provide specific information that would tell farmers what to do. Farmers need to know harvest times, irrigation timing, when to plant certain crops, or when in the winter will the crops freeze.

Q2: AI weather systems use many different data sources. Name and briefly explain at least FOUR types of data that an AI farm weather system collects and uses. For each, explain WHY that specific data type is useful for a wheat farmer in West Texas.

A2: Historical rainfall - how much rain the area receives generally. This is useful to a wheat farmer as it gives them a general time of when it will rain in the area. Local temperature - the ambient temperature of the area. This lets the farmer know when the optimal temperature occurs so that they can plant the wheat. Wind speeds - how intense the wind is and what direction it comes from. This allows the farmer to plant the wheat accordingly to avoid unwanted seed distribution from wind gusts. Crop growth stages - where the wheat is in its life cycle. This is probably the most useful for a wheat farmer, as it tells them when to harvest or when to fertilize the crops.

Q3: The Torres family pays \$1,500/year for the AI weather service. Last year, they lost \$54,000 (30% of \$180,000 crop value) because a frost was missed. If the AI weather system could predict late frosts with 85% accuracy and they could protect the crop (e.g., with wind machines or timing their planting differently): How much money could they potentially save? Is \$1,500/year a good investment? Show your calculation.

A3: $0.85 * \$54,000 - \$1,500 = \$44,400$ saved. \$1,500 a year is a decent investment, as they are losing less crops and therefore less money each year with the new AI system.

Q4: No AI system is 100% accurate. Describe TWO scenarios where the AI weather prediction could be WRONG and explain the consequences for the Torres family in each case. Then explain why human judgment (the farmer's experience and local knowledge) should still play a role even when using AI.

A4: The model could incorrectly predict when a storm will come, which will cause the family to prepare for a storm that will not come. The AI could also falsely claim that the crops are ready for harvest when they are still growing, which could lead to a premature harvest that would cost the family a lot of money. Human

judgement, such as family traditions/experiences with farming, is still something that an AI cannot replicate.

Q5: The Ogallala Aquifer (which the Torres family uses for irrigation) is being depleted. Explain how AI weather prediction could help the Torres family use significantly less water for irrigation. Give at least TWO specific examples of how a weather AI would change their irrigation decisions. Research: how much water does traditional vs. AI-guided irrigation typically save?

A5: The model could predict rain cycles and soil moisture levels during certain days to limit irrigation usage. Precise irrigation usually saves farmers 30-50% more water, which adds up to thousands of gallons per seasonal cycle.

Q6: Using the scenario below, create a 14-day farm planning table for the Torres family. Imagine the AI has given them the following forecast for the first two weeks of April (planting season): Days 1-3: Sunny, 24C, no rain. Days 4-6: Cold front arriving, temperature dropping to 5C, frost risk HIGH. Days 7-9: 60% chance of rain, moderate. Days 10-12: Clear, 22C, wind 20km/h. Days 13-14: Possible hail. For each period, write what farming action the Torres family should take and WHY the AI data changes their decision.

A6:

Day(s)	Weather Forecast	Recommended Action	Why the AI Forecast Changes the Decision
1–3	Sunny, 24°C, no rain	Early soil prep & initial planting — tilling, compost spreading, planting hardy crops (potatoes, onions, leafy greens). Begin irrigation lightly.	Warm, dry conditions allow soil to be worked easily. Seeds germinate well at 20–25°C. No rain means irrigation must be controlled manually.

4–6	Cold front, 5°C, HIGH frost risk	Pause planting & protect seedlings — cover crops with frost cloth, move trays indoors, add mulch around early plantings. Avoid watering at night.	Frost can kill young seedlings and damage tender leaves. Planting now would waste seeds and labor. Protection measures reduce freeze damage.
7–9	60% chance of moderate rain	Resume planting & use rainfall for germination — plant moisture-loving crops (beans, corn). Apply organic fertilizer before rain. Check drainage.	Rain provides free irrigation and helps seeds germinate. But moderate rain can cause runoff, so drainage checks prevent waterlogging.
10–12	Clear, 22°C, wind 20 km/h	Field maintenance & pest control — stake young plants, prune damaged leaves, apply wind-safe foliar feed. Avoid spraying pesticides during peak wind.	Wind can break stems and reduce spray effectiveness. Stable temperatures allow safe outdoor work but require securing plants.
13–14	Possible hail	Crop protection & equipment sheltering — cover vulnerable crops with row covers, move tools/tractors under shelter, delay any harvesting. Inspect roofs and gutters.	Hail can shred leaves, bruise fruit, and damage equipment. Protective covers reduce impact damage; delaying harvest prevents crop loss.

Reflection Q1: Food Security and Climate Change: Climate change is making weather MORE unpredictable — more extreme droughts, unexpected frosts, heavier storms. How does AI weather prediction become MORE important (not less) as climate change continues? Give one specific example of how AI could help a farmer adapt to a climate change impact.

Reflection A1: AI prediction of weather events as climate change occurs is more important now than ever. These models can aid farmers by analyzing the changing weather patterns and predict these changes for future reference. Farmers could then use this model and adapt their farms based on the predictions.

Reflection Q2: The Digital Divide in Farming: The AI weather service costs \$1,500/year. A large corporate farm with 50,000 acres finds this very affordable. A smallholder farmer in rural Kenya with 2 acres finds it completely unaffordable. What are the ethical implications of AI making some farmers much more competitive while others cannot access it? What could be done to make AI weather tools accessible to small farmers globally?

Reflection A2: Those who cannot afford an AI weather service are going to be at a competitive disadvantage, since they will not have the advanced technology that will aid them in their agriculture practices. In order to make these technologies more accessible to farmers globally, the company can either reduce the price significantly, or deploy large-scale models that will oversee multiple farms at a reduced price.

Reflection Q3: Farmer Knowledge vs. AI: A third-generation farmer like Mr. Torres has decades of experience reading the land — he notices things like how his cattle behave before a storm, or how the dust smells when rain is coming. Do you think AI should REPLACE this kind of traditional farming knowledge, or COMPLEMENT it? Give a specific example of when human experience would catch something the AI might miss.

Reflection A3: I believe that AI should not replace these traditions. I think that AI should complement this knowledge of farming, or even enhance them further. A specific scenario where human experience would outweigh an AI's abilities would

be tornado detection in Texas. Some people can tell a tornado is coming based on the color of the sky during a storm or when a storm is approaching. An AI would solely rely on weather data and not human instinct.